

1.1 Propositional Logic - 2

Tuesday, September 3, 2024

11:00 AM

If a proposition can be expressed using $\wedge, \vee, \oplus$ it is called a compound proposition. Otherwise it is called an atomic proposition.

Definition: The conditional statement $P \Rightarrow Q$ is the proposition "if P , then Q ." The statement is only false when P is true and Q is false.

Truth Table For $P \Rightarrow Q$

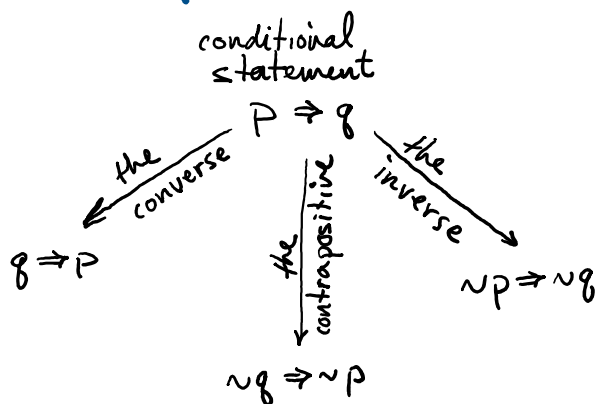
P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

This is also called an implication.

$$P \Rightarrow Q$$

hypothesis $\Rightarrow$ conclusion

Good list of ways implication may appear on page 7.



Definition: Two conditional statements are equivalent if they have the same truth tables.

P	Q	$P \Rightarrow Q$	$\sim Q \Rightarrow \sim P$
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P	q	$P \Rightarrow q$	$\sim q \Rightarrow \sim P$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

P	q	$q \Rightarrow P$	$\sim P \Rightarrow \sim q$
T	T	T	T
T	F	T	T
F	T	F	F
F	F	T	T

Based on these truth tables we see that

$P \Rightarrow q$ is equivalent to $\sim q \Rightarrow \sim P$

$q \Rightarrow P$ is equivalent to $\sim P \Rightarrow \sim q$

Definition: A biconditional statement is denoted $P \Leftrightarrow q$ and is true only when P and q have the same truth values.

Truth Table For $P \Leftrightarrow q$

P	q	$P \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

This is often written by inserting the phrase "if and only if" between the propositions P and q . Sometimes abbreviated to "iff".

The biconditional is equivalent to conjunction

$$P \Leftrightarrow q \cong (P \Rightarrow q) \wedge (q \Rightarrow P)$$

With our logical operators and conditional

With our logical operators and conditional operators, we can analyze the truth values of complex statements using larger truth tables.

Construct the truth table for the compound proposition $\sim q \vee (\sim p \wedge q)$

P	q	$\sim q$	$\sim p$	$\sim p \wedge q$	$\sim q \vee (\sim p \wedge q)$
T	T	F	F	F	F
T	F	T	F	F	T
F	T	F	T	T	T
F	F	T	T	F	T

In general, parentheses will indicate the order of operations but if the order is ever unclear, they are performed in the following order.

